

SEMESTER EXAMINATION-2021	
CLASS – B.TECH III SEMESTER	
SUBJECT : DIGITAL SYSTEM DESIGN	
PAPER CODE: BET-C306	
Time: 3 hour	Max. Marks: 70 Min. Pass: 40%
Note: Question Paper is divided into two sections: A and B . Attempt both the sections as per given instructions.	
SECTION-A (SHORT ANSWER TYPE QUESTIONS)	
Instructions: Answer any five questions in about 150 words each. Each question carries six marks.	(5 X 6 = 30 Marks)
Question-1: Obtain the truth table of following function: $F = xy'z + x'y'z + w'xy + wx'y + wxy$	
Question-2: Simplify the following Boolean function by first finding the essential prime implicants: $F(A,B,C,D) = \sum (0,2,3,5,7,8,10,11,14,15)$	
Question-3: Design a combinational circuit with three inputs that generates 1's complement of input binary.	
Question-4: Convert SR Flip-Flop to D Flip-Flop.	
Question-5: Design a BCD asynchronous counter using JK flip-flop.	
Question-6: With the help of neat diagram, explain the working of Bidirectional shift register.	
Question-7: Implement the full subtractor using decoder and OR gates.	
Question-8: With the help of neat diagram, explain the working of CMOS inverter.	
Question-9: Represent the number $(A6)_{16}$ in decimal number system. Also find its BCD code, Gray code and Excess -3 code.	
Question-10: Perform following operation : I. Add -25 to -75 using 2's compliment. II. Add 256 to 569 in BCD code.	

SECTION-B (LONG ANSWER TYPE QUESTIONS)	
Instructions: Answer any FOUR questions in detail. Each question carries 10 marks.	(4 X 10 = 40 Marks)
Question-11: Reduce the following Boolean expression by using K-Map and implement the reduced function by Using NAND gates. $F(A,B,C,D) = \pi(1, 3, 5, 7, 13, 15)$	
Question-12: Design a 3 input Gray to Binary code converter by using Decoder.	
Question-13(a) : Design a BCD adder by using IC 7483. (b). Design a 4-bit magnitude comparator. Also explain the functioning of IC 7485.	
Question-14: Implement the following logic function by using 8 : 1 MUX. Take D as input variable. $F(A,B,C,D) = \sum(1,2,4,5,7,9,12,13,15)$	
Question-15: Describe fan-in, fan-out, noise margin, Threshold voltage, propagation delay, power dissipation and speed power product in reference to logic families.	
Question-16 (a) : With the help of a neat diagram, explain the working of CMOS NOR gate. (b). With the help of a neat diagram, explain the working of 2 input TTL NAND gate.	
Question-17: A sequential circuit has two JK flip-flops A and B and one input x . The circuit is described by the following flip-flop input equations: $J_A = \bar{X}, \quad K_A = \bar{B}$ $J_B = X, \quad K_B = A$ (a). Derive the state equations A (t + 1) and B (t + 1) by substituting the input equations for the J and K variables. (b). Draw the state diagram of the circuit.	
Question-18: Design a synchronous counter using D-FF with the following repeated binary sequence: 0, 1, 3, 4, 7. Check whether counter is self-start or not. If not then make the solution.	